

Jake Zheng

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Education

The University of Chicago

June 2028 (expected)

B.A. Statistics, B.A. Mathematics, B.S. Computer Science | GPA: 3.77/4.0

Chicago, IL

- **Graduate Coursework:** Machine Learning, Mathematical Statistics, Modern Methods in Applied Statistics, Convex Optimization, Applied Linear Regression, Robotics & Reinforcement Learning
- **Undergraduate Coursework:** Mathematical Probability, Real Analysis, Abstract Linear Algebra, Honors Economic Analysis, Theory of Algorithms, Discrete Mathematics, Systems Programming

Thomas Jefferson High School for Science and Technology

June 2024

High School Diploma

Alexandria, VA

Experience

Fermi National Accelerator Laboratory (DUNE)

June 2026 – Aug. 2026

AI/ML Research Intern

Batavia, IL

Wasserstein Normalized Autoencoder (WNAE)

- Engineered an unsupervised WNAE under the supervision of Dr. Seon Hee Seo to identify anomalous neutrino events as proxies for Beyond Standard Model (BSM) physics at the ProtoDUNE Vertical-Drift (NP02) detector based at CERN.
- Optimized and fine-tuned a normalized Boltzmann energy model via finite-step Langevin dynamics and an empirical sliced 2-Wasserstein optimal-transport loss function to steady convergence.
- Designed an 11.2K-parameter convolutional architecture featuring a custom charge-aware ℓ_1 reconstruction-energy penalty to isolate sparse spatial tracks and eliminate background noise.
- Demonstrated that the WNAE uniquely recovered 20.8% of anomalous neutrino-proxy events completely missed by the baseline ADCSimpleWindow (ADCSW) trigger at a 5 Hz operational threshold.
- Presented findings to 50+ scientists at DUNE AI/ML conference.

ConvNeXt V2 Masked Autoencoder (MAE)

- Developed a MinkowskiEngine-based dense convolutional MAE for simulated ProtoDUNE data, adapted from Meta FAIR.
- Designed a 1.5M-parameter convolutional architecture using an ℓ_2 loss to learn sparse spatial tracks and eliminate background noise.
- Demonstrated that the MAE uniquely recovered 23.5% of anomalous neutrino-proxy events missed by the baseline trigger rate

Code may be available by request

Publications

Zheng, J. (2026). *Wasserstein normalized autoencoder for anomaly detection in ProtoDUNE vertical-drift detector* (Report No. FERMILAB-PUB-26-0510-STUDENT). Fermi National Accelerator Laboratory. <https://inspirehep.net/literature/3193146>

Projects

Black Box Variational Inference for Non-Conjugate Preference Models

May 2026 – Present

- Designed a PyTorch BBVI framework to approximate non-conjugate posteriors for a joint Bradley-Terry preference model and Poisson model to extract features from Last.fm dataset.
- Implemented Rao-Blackwellization and Leave-One-Out (LOO) control variates into the ELBO optimization to isolate local signals, eliminate gradient noise, and stabilize trajectory convergence.
- Recovered true latent parameters with a 0.9978 Pearson correlation while identifying feature clusters.

Hidden Markov Models for USS Scorpion Submarine Spatial Tracking

March 2026 – Present

- Vectorized Forward-Backward inference in NumPy to optimize runtime across an 858-cell grid.
- Modeled spatial transitions for 6 latent scenarios conditioned on a binomial time-of-death distribution.
- Applied exact variable elimination to Poisson sonar data to compute posterior smoothing probabilities.

Leadership

Maroon Capital Quant Finance Club | *Quant Analyst*

Sep. 2025 – Present

- Developed a Python/Alpaca pairs trading strategy (PM/MNST) for a derivatives pricing capstone.

Financial Markets Program (FM) | *Member*

Feb. 2025 – Present

- Selective, 3-year program focused on quantitative finance at the Chicago Booth School of Business.

Asian American InterVarsity (AAIV) | *Small Group Leader*

Sep. 2025 – Present

Skills

Languages: Python (PyTorch, TensorFlow, Keras, NumPy, Pandas), C, C++, SQL

Interests: Competitive Valorant (Top 0.1% Global Rank, 22x Seasons), Acoustic Guitar